

B.E. I Yr. (E&TC – A & B section)
Mid Test-II Date- 10/01/2023
Sub: Electrical Engineering (2EIRC4)

Time: 70 min

Mark: 20

Note: All questions carry Equal marks. Assume suitable data if required.

1. Explain B-H curve with along with Coercive force and residual magnetism and what is its significance?
2. A steel ring of 25 cm mean diameter and of circular section 3cm in diameter has an air gap of 1.5 mm length. It is wound uniformly with 700 turns of wire carrying a current of 2 A. Calculate (1) MMF (2) flux density (3) magnetic flux (4) reluctance (5) Relative permeability of the ring. Assume that iron path takes 35% of the total MMF.
3. A 200 KVA, 11000/230 V, single phase transformer gave following test results: No load test at rated voltage $W_0 = 1600W$,
Short circuit test at rated current, $W_{sc} = 2600W$. Calculate its all day efficiency with below load cycle:
200KVA, 0.8pf, 8hours
100KVA, unity pf, 6 hours
No load, 10 hours.
4. A resistance of 20Ω and inductance of $0.2H$ and capacitance of $100\mu F$ are connected in series across 220V, 50Hz mains. Determine the following: Impedance, current, voltage across R,L,C, power in watts and p.f. and angle of lag.

Tapeesh sir

B.E. I Yr. (Common to IT A, IT B, Civil)

Mid Test-II Date-30/05/2023

Sub: Electrical Engineering (2EIRC4)

Time: 70 min

Mark: 20

Note: All questions carry Equal marks. Assume suitable data if required.

1. Write Short note on (1) B-H curve along with its significance (2) Auto Transformer.
2. A steel ring of 25 cm mean diameter and of circular section 3cm in diameter has an air gap of 1.5 mm length. It is wound uniformly with 700 turns of wire carrying a current of 2 A. Calculate (1) MMF (2) Flux density (3) magnetic flux (4) reluctance. (5) Relative permeability of the ring. Assume that iron path takes 35% of the total MMF.
3. A 200 KVA, 11000/230 V, single phase transformer gave following test results: No load test at rated voltage $W_o = 1600$ W,
Short circuit test at rated current, $W_{sc} = 2600$ W. Calculate (1) all day efficiency with below load cycle:
200KVA, 0.8pf, 8hours
100KVA, unity pf, 6 hours
No load, 10 hours.
4. A parallel circuit has two branches. One branch contains a resistance of 8 ohm and inductance of 19.1mH in series and the other branch contains a resistance of 6 ohm and capacitor of capacitance 601.55 micro farad in series. This parallel circuit is connected across a supply voltage of 240 V, 50Hz. Determine (1) Current drawn by each branch (2) total current drawn by each branch (3) Power factor of whole circuit. Solve using admittance method only.